

Graph Neural Network Modeling of Breast MRI Vascular Networks for Treatment Response Prediction

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Introduction + Motivation

Breast cancer is the most common cancer in women worldwide, and neoadjuvant chemotherapy (NAC) is widely given before surgery. Only about a third of patients achieve a **pathologic complete response (pCR)** to NAC, which is associated with improved survival outcomes. NAC is an intensive treatment that can have significant side effects, and patients who do not achieve pCR may require additional treatment. We would like to develop methods to predict which patients will respond to NAC, so that we can better tailor treatment plans and improve outcomes.

Ultrafast MRI Dataset

The dataset consists of **ultrafast breast MRI** from a single institution: 181 labeled exams across 143 patients, of which 179 are analyzed here. Each exam is a dynamic contrast-enhanced series with precontrast baseline frames and pCR labels; 35% of exams are pCR-positive. Exams come from three sub-sources (**simbiosys**, **uch_nac**, **her2_naclike**) whose acquisition timing differs materially — 13–24 frames over 96–480 s.

Graph Representation

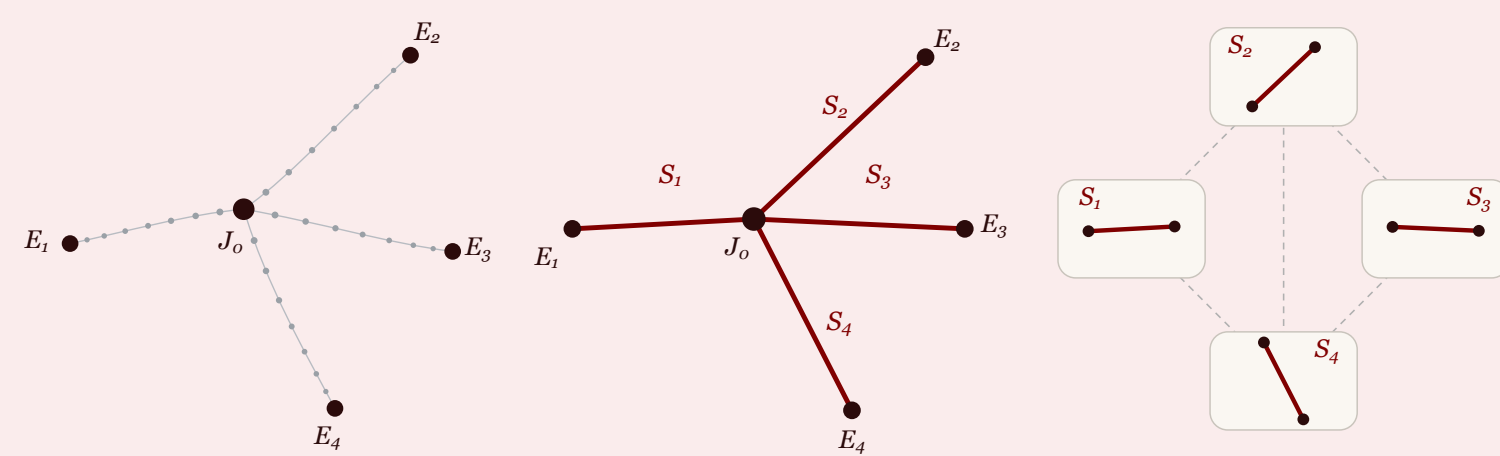


Figure 1. Vascular segments and junctions are converted into graph nodes and edges.

We represent the vascular network of each patient's tumor as a graph, and propose 3 different methods of constructing the graph from the MRI scans. In the following let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph.

- **Voxel-as-node** Let the set of nodes $\mathcal{V}_{\text{vox}}$ be the set of all voxels, and let the set of edges $\mathcal{E}_{\text{vox}}$ be the set of all pairs of adjacent voxels. In this representation, each voxel is a node, and edges connect adjacent voxels.
- **Junction-as-node** Let $\mathcal{V}_{\text{jun}}$ be the set of all junctions (i.e. $e \in \mathcal{E}_{\text{vox}}$ with $\deg(e) > 2$) in the vascular network, and let $\mathcal{E}_{\text{jun}}$ be the set of all pairs of junctions that are connected by a vascular segment.
- **Segment-as-node** Let $\mathcal{V}_{\text{seg}}$ be the set of all vascular segments, and let $\mathcal{E}_{\text{seg}}$ be the set of all pairs of segments that are connected at their endpoints. In this representation, each segment is a node, and edges connect adjacent segments.

Prediction Pipeline and GNN Architecture



Figure 2. End-to-end prediction pipeline.

Vessel segmentation is performed using a **nnU-Net model**, which labels each voxel as vessel or background. The segmentation is simplified into a one-voxel wide network of vessels using the **tc4d** algorithm, then converted into a graph using one of the three representations above.

A **graph neural network (GNN)** predicts treatment response from that graph. Successive **graph convolution (GCN)** layers aggregate information from neighboring nodes, a global pooling layer collapses all nodes into a fixed-size representation of the entire graph, and a fully connected layer maps that representation to a binary prediction (pCR vs. non-pCR).

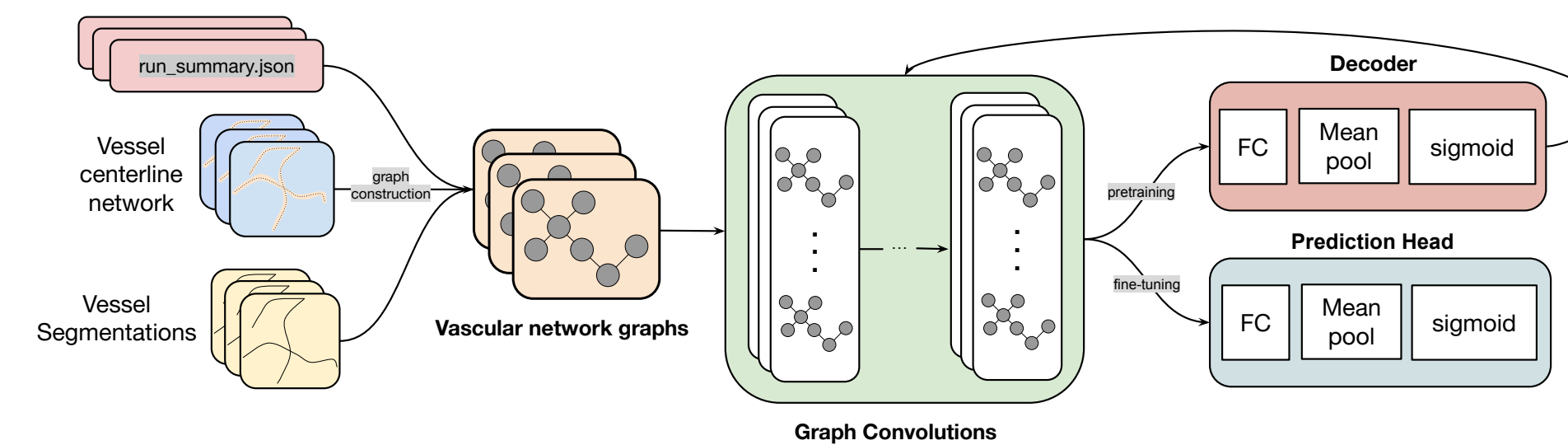


Figure 3. GNN architecture: 4D DCE-MRI through the vessel graph to a pCR probability.

Self-supervised Pretraining

To improve the performance of the GNN, we use a **self-supervised pretraining** approach. The GNN is first trained to predict *per node* the future contrast enhancement, where it learns to predict contrast flow through the network.

Each contrast curve is cut into non-overlapping windows. Shown the frames on an input horizon $[t_1, t_L]$, the model predicts every node's enhancement over the following horizon $[t_L, t_{L+H}]$, scored by mean absolute error against the measured signal. Predictions are conditioned on **elapsed seconds** $\tau = t - t_1$ rather than on frame index, so the irregular ultrafast cadence is handled directly.

Results

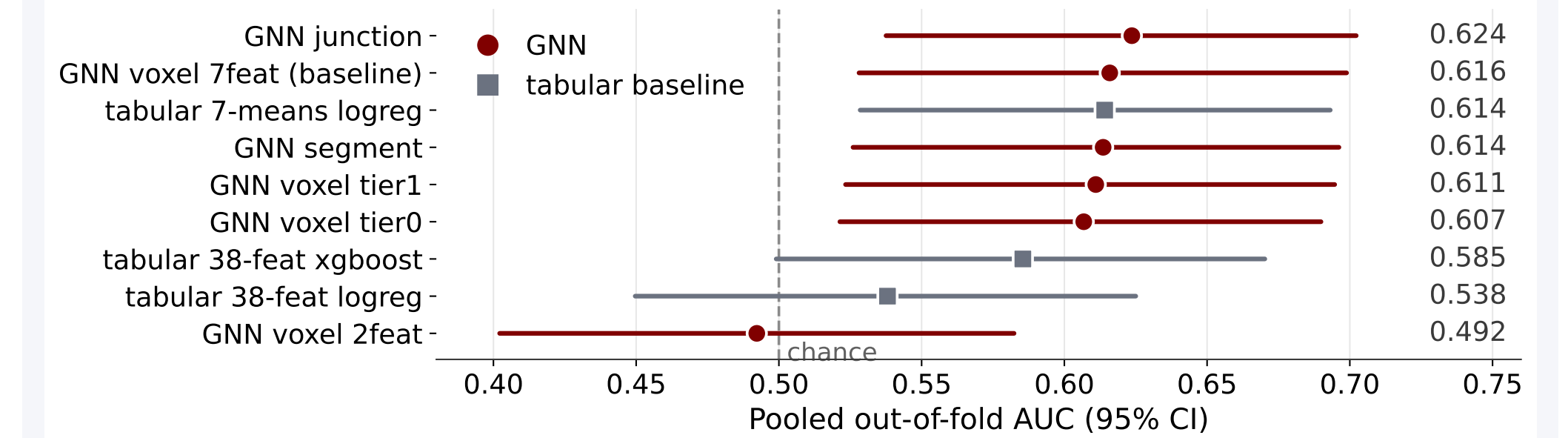


Figure 4. Pooled out-of-fold AUC, bootstrap 95% CI, shared folds. *Preliminary.*

Every arm carrying the full enhancement-kinetic feature set clusters between **0.607 and 0.624** AUC. The paired difference between the best GNN and the best tabular baseline is consistent with zero, $+0.002 [-0.041, 0.046]$.

A GNN restricted to two features (peak time and radius) sits at chance, 0.492, placing the signal in the enhancement kinetics rather than in the graph. So far feature content moves the metric; model class and graph representation do not.

Protocol confound

Acquisition timing differs by sub-source, and four of the seven kinetic features depend on it. A protocol-only model with no imaging scores 0.593; regressing protocol out per fold leaves 0.539 — read the vessel signal as the bracket $[0.539, 0.613]$.

Reproducibility

All code is publicly available on GitHub: vessel segmentation, graph construction, GNN training and evaluation, plus instructions to reproduce these results.

Conclusion

We built an end-to-end pipeline that turns ultrafast breast MRI into a vessel graph and predicts pCR with a GNN. The vasculature carries some signal, but no graph representation yet beats a tabular summary of the same node features, and part of the pooled signal tracks acquisition protocol rather than physiology. Quantifying what survives protocol adjustment, and whether pretraining helps, are the next steps.

References

- [1] Siyi Tang, Jared A. Dunnmon, Khaled Saab, Xuan Zhang, Qianying Huang, Florian Dubost, Daniel L. Rubin, and Christopher Lee-Messer. Self-supervised graph neural networks for improved electroencephalographic seizure analysis, 2022. URL <https://arxiv.org/abs/2104.08336>.